



UNIVERSITY OF SANTO TOMAS
COLLEGE OF SCIENCE
Academic Year 2026 - 2027



DSA4157 - BUSINESS INTELLIGENCE

1ST TERM AY 2026-2027

Activity #1: Adventure Works BI Solutions (Challenge)

McReg Analytics

“We are speed. We are McReg. KAAA-CHOW!”

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ADVENTURE WORKS PERFORMANCE ANALYTICS DASHBOARD

BUSINESS CASE

By McReg Analytics

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I. BACKGROUND

Adventure Works Cycles, a fictional company from the free database *AdventureWorks* provided by Microsoft, is determined to continue its success as a manufacturer and distributor of bicycle parts and accessories by selling its products through different channels: directly to primary consumers by Internet Sales and to other businesses by Reseller Sales—encompassing wholesale offers and distribution from its pre-existing network of resellers and distributors.

While the dataset, *AdventureWorksDW2025*, consists of transactional data, mainly customer and employee information, product categories, marketing promotions and discounts, reseller networks, sales geographic locations, and consolidated information on reseller and internet sales, along with their combined sales quota.

To support Adventure Works Cycles in translating this data into actionable business direction, our data consulting team, McReg Analytics, is determined to explore the data, design a dashboard, and deliver a business intelligence solution that consolidates performance across channels, regions, and product lines into a single, reliable source of truth for company leadership.

II. PROBLEM STATEMENT

Despite the company *Adventure Works Cycles* utilizing two different channels for marketing and distribution—Internet and Reseller Sales—and providing marketing promotions and discounts, its sales performance has only been monitored through static reports, which makes it difficult for the business, marketing, sales, and data strategists and analysts from the company to identify underperforming territories, compare and evaluate the effectiveness of implementing both internet and reseller sales, and assess whether the promotional discounts have been financially and operationally effective for the company in real-time. There is an apparent need for a live dashboard so that Adventure Works Cycles can swiftly identify potential risks and ensure positive growth through making data-driven decisions.

III. STAKEHOLDERS

The primary stakeholders are the Chief Business Officer and the Chief Operating Officer of Adventure Works, who are responsible for making monthly and quarterly financial and operational decisions, whether it be overseeing the company's financial health or managing the company's overall direction in business operations, including sales, logistics, and marketing.

The secondary and tertiary stakeholders are: the Director of Sales, who is responsible for identifying underperforming territories and monitoring the performance of the sales and distribution channels; the Director of Marketing, who is responsible for evaluating the effectiveness of promotions and discounts and assess whether profit and volume are proportionally increasing; and the Head of Data Science and Analytics to ensure that the models and dashboards are consistent, accurate, and reliably ready for deeper analysis as these metrics are vital in ensuring the integrity of what is being shown and reported.



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IV. KEY BUSINESS QUESTIONS

- What are our total revenue, gross profit, and margin trends across B2B (Reseller) and B2C (Internet) channels?
- Are promotional discounts generating incremental sales volume, or are customers buying flagship products at full price regardless?
- How can we reconfigure pricing tiers and reallocate production capacity away from low-margin lines to maximize gross margin?
- Which customer and channel segments offer the highest potential for bundling high-margin accessories and apparel with core bike sales?

V. ANTICIPATED BUSINESS VALUE

AdventureWorks has a major opportunity to boost their sales and profits by simply changing how they price their products and how it's being sold. As we delve deep into their data more, sales show that customers are more than happy to pay the full price even on high-end bikes. Additionally, the promotional discounts on these products does not help boost their sales, it just cuts their profit even more. By stopping unneeded discounts and making small price updates to high-demand bikes, AdventureWorks can immediately protect its bottom line and keep more profit from every sale.

Another key step is getting customers to buy smaller, highly profitable items alongside their bikes. Bicycles are expensive to build, which keeps their profit margins naturally low. However, items like helmets, water bottles, tires, and apparel have very high profit margins. Bundling these high-margin accessories with bike purchases makes every customer transaction much more profitable overall, while shifting factory production toward the bike models that generate the highest returns.

VI. SUCCESS MEASURES AND EVALUATION CRITERIA

Financially, our primary indicator is gross profit margin percentage expansion, which tracks overall profitability gains beyond our historical baseline as full-price sales replace discounted volume. To ensure we capture maximum value per transaction, we will measure average realized price per unit alongside our full-price revenue ratio to verify that flagship bike series convert at list price without reliance on promotions. Additionally, we will evaluate high-margin attach rates to track how effectively accessories and components are bundled with primary bike orders, while monitoring wholesale account margin stability to prevent profit leakage. On a strategic level, the framework evaluates how effectively operational changes protect top-line growth while optimizing margins. We will monitor price elasticity and sales volume retainage to confirm that modest price adjustments on high-volume, low-margin categories like road bikes maintain steady order volumes. Success will also be measured by our product mix shift, ensuring factory capacity moves away from lower-margin lines like touring bikes toward higher-margin models.